

LLM-Based Software Engineering Research Paper Replication and Enhancement

Problem Statement

Large Language Models (LLMs) are increasingly being used in various Software Engineering tasks such as Requirement Engineering, Code Generation, Testing, Bug Detection, Documentation, and UML Generation. Many recent research papers propose AI-driven approaches for solving software engineering problems, but reproducing and validating these results is an important part of research and innovation.

Select any research paper available online that combines Software Engineering and Large Language Models (LLMs). Your task is to replicate the proposed work, evaluate its results, and introduce a novel enhancement or improvement over the original approach.

The project should demonstrate both reproduction of the research work and contribution through additional innovation.

Objective

The objective of this assignment is to:

- Understand current research trends in Software Engineering with LLMs
- Learn how to read and analyze research papers
- Replicate existing AI-based Software Engineering research
- Evaluate experimental results and methodologies
- Introduce novelty or improvement to existing research
- Develop research-oriented implementation and evaluation skills

Features to Implement

- Select a research paper related to Software Engineering and LLMs
- Study and understand the proposed methodology
- Replicate the implementation or workflow presented in the paper
- Reproduce experiments and results
- Compare reproduced results with the original paper
- Identify limitations or gaps in the existing approach
- Add novelty or enhancement to the proposed system

- Evaluate the impact of the proposed enhancement
- Generate proper experimental analysis and comparisons
- Document methodology, implementation, results, and improvements
- Present findings with graphs, tables, or evaluation metrics
- (Bonus) Publish code on GitHub or prepare a short research-style report/paper on the enhanced work